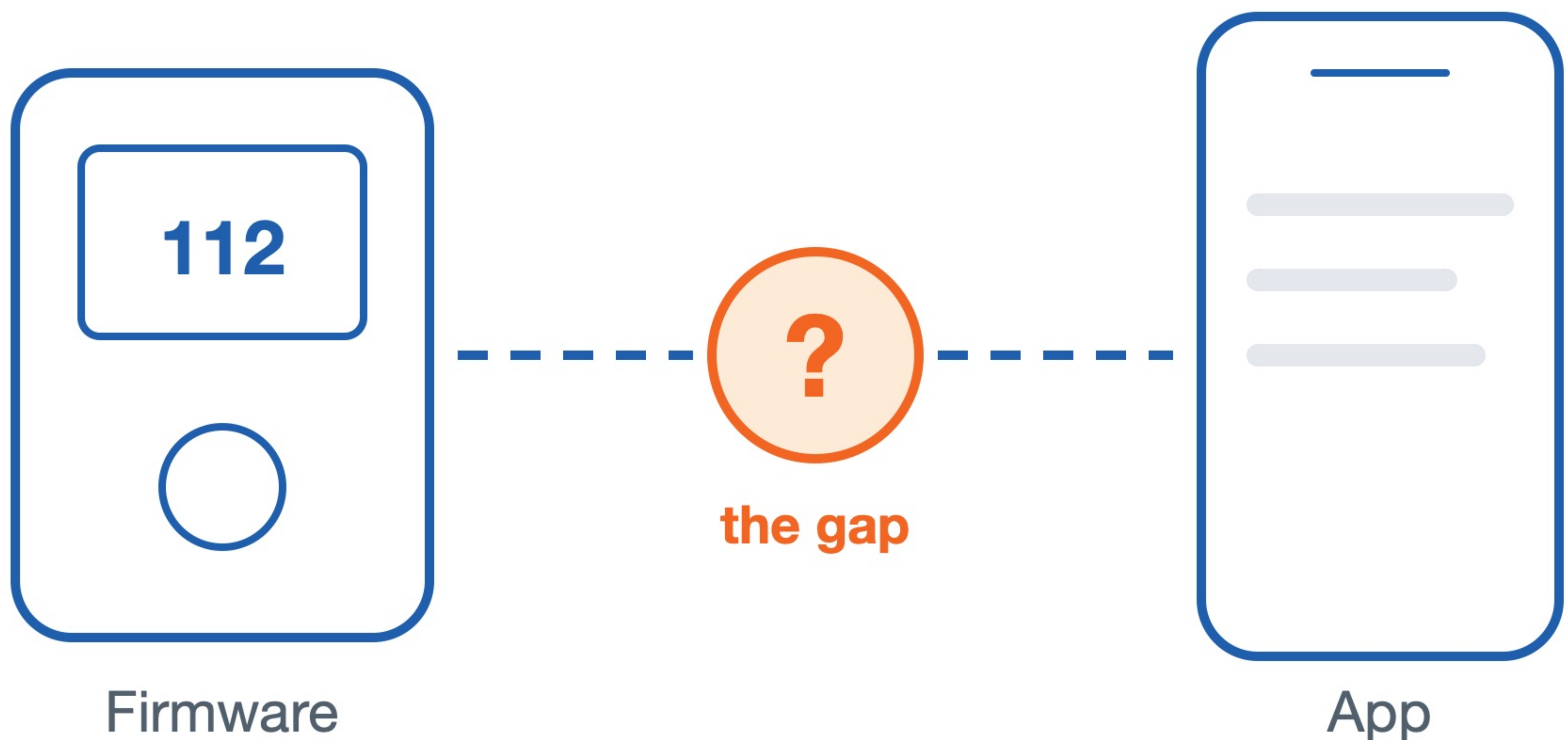


A BLE MEDICAL DEVICE PLAYBOOK

The meter works.

The app works.

The reading still goes missing.



5 things to lock before
either team writes code.

One product. Two teams.

Firmware passes its tests.

The app passes its tests.

The failures live in the gap between them:
the BLE link, the data and the timing.

FROM OUR WORK

We built BLE glucometer firmware
and Android + iOS apps for a US
healthcare organisation.

These are the 5 things we settle first.



WHERE THE DATA LIVES

Delete only after the app says yes.

Delete a reading on send, and one dropped connection loses it for good. Keep it on the meter with a sequence number. The app asks for every record since the last one it saved.

The Bluetooth SIG Glucose Profile does this with its Record Access Control Point (RACP).

AGREE ON

Storage size, record format, and who deletes what, and when.



Decide whose clock is right.

A reading with the wrong time misleads anyone reviewing the trend.

The meter stamps each reading. The phone corrects the clock on connect.

Battery swaps and time-zone travel are the normal case, not the edge case.

AGREE ON

Which clock wins, and how a correction shows up in the history.



POWER

Choose how the meter sleeps.

Between readings the meter only needs to be findable. Low-power advertising does that, but the interval is a trade: longer saves battery, and the patient waits longer for the app to connect.

BLE allows advertising intervals from 20 ms to 10.24 s.

AGREE ON

The battery-life target, and the wait a patient will accept after a test.

4

THE PHONE

Design for what the phone allows.

iOS suspends apps in the background.
Android 12+ asks for new Bluetooth
scan and connect permissions.

A patient who never opens the app
still expects the reading to arrive.
So does their new phone.

AGREE ON

What syncs in the background, and how
a meter re-pairs with a new phone.



VERSIONS

Old firmware meets the new app.

The app updates through the store in days. Meter firmware may never update at all.

So every app release has to talk to every firmware version still in use.

AGREE ON

A versioned protocol, and an app × firmware test matrix for each release.

Give the gap one owner.

Write the five down as one shared spec,
not two team documents.

- 1 Where a reading lives until confirmed
- 2 Whose clock is right
- 3 How the meter sleeps
- 4 What the phone allows
- 5 Which versions must talk

When one team owns meter, app and link,
the gap has an owner too.

Follow CoreFragment for more